Supplementary Material: A Critical Survey of Research in Music Genre Recognition (ISMIR 2024)

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Citation numbers follow those in the main text.

S1 Methodology

S1.1 Using Google Scholar

We chose to use Google Scholar as our main data source because it seemed to detect a significant number of publications not accounted for in more curated collections like Web of Science or Scopus. This better serves our aim for comprehensiveness ahead of any value judgments about the quality of a publication, but also entails some trade-offs. Using Google Scholar in this way requires considerably greater manual effort in sifting and entering results, checking data and, sometimes, obtaining PDFs.[1]

S1.2 Criteria for Inclusion

In the compilation of our collection of publications, we make a distinction between "genre" and "style". For instance, we do not include

> W. Herlands *et al.*, "A machine learning approach to musically meaningful homogeneous style classification," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 28, no. 1, 1 Jun. 19, 2014

since it is about identifying the composer of a work. We do not include

> H. V. Koops *et al.*, "Corpus-based rhythmic pattern analysis of ragtime syncopation," in *Proceedings of the 16th International Society for Music Information Retrieval Conference*, ISMIR press, Oct. 26, 2015, pp. 483–489

because it is a computational analysis of syncopation in piano music of one style (ragtime). We also do not include short papers resulting from MIREX challenges,[2] or bachelors' theses or student course projects. We do include masters' theses, however. Finally, we exclude publications where genre is not recognized by machines, such as

> F. A. Raposo *et al.*, "Learning Low-Dimensional Semantics for Music and Language via Multi-Subject fMRI," *Neuroinformatics*, vol. 20, no. 2, pp. 451–461, Apr. 1, 2022.

S1.3 Publication Type

Some references can be assigned to two or more of these types, e.g., [235, 236] builds MGR systems and then sees how sensitive they are to adversarial examples, thereby criticizing standard MGR evaluation methodology. And Sturm [471] surveys all experimental MGR work using the GTZAN dataset, identifies several faults in the dataset, and then trains and tests several MGR systems in the dataset under different conditions.

[1] Because Google Scholar doesn't have an API, this work is most practically done in the browser. The Zotero browser plugin—which allows one to input multiple sources with a single click— makes this substantially less arduous than it might otherwise be. However, one must be careful not to go too quickly, in case Google Scholar decides you are a 'bot' and locks you out.

[2] https://www.music-ir.org/mirex/wiki/MIREX_HOME (Last access Apr. 9 2024

S1.4 Dataset

Where experiments have used fine-tuning or transfer-learning, we record only the target data sets used in experiments and not the data sets used for pre-training.

S1.5 Experimental Design

The experimental design categories described by Sturm [478, p. 32] are: *Classify*, *Features*, *Generalize*, *Robust*, *Scale*, *Cluster*, *Retrieve*, *Rules*, *Compose*, *Eyeball*. Briefly, with an example publication given for each, the *Classify* experimental design involves having a system reproduce the ground truth labels of a dataset, e.g., [451]. If different features are tested, then its design is *Features*, e.g., [405]. If several different datasets are used, then the design is *Generalize*, e.g., [56]. If experiments involve adding noise or other distortions to an input, then the design is *Robust*, e.g., [405]. If experiments involve tests with varying numbers of genres in the same dataset, then its design is *Scale*, e.g., [259]. If a publication shows attempts at clustering data, then its design is *Cluster*, e.g., [331]. If an experiment involves retrieval of music according to characteristics of an input, then its design is *Retrieve*, e.g., [427]. If the researcher attempts to uncover the rules used by an MGR system, then its design is *Rules*, e.g., [213]. If the researcher uses a system to generate music in the styles it has learned, its design is *Compose*, e.g., [480]. Finally, if the researcher compares plots characterizing style or genre, then its design is *Eyeball*, e.g., [551]. Some experiments involve many different designs, e.g., Kim et al. [240] employ Classify, Feature, Generalize, Robust, Cluster, and Eyeball.

S1.6 Figure of Merit (FoM)

For a set of labeled items submitted to a music genre recognition system, *accuracy* is the proportion of those the system has labeled the same way. For a set of test items of a given label, *recall* is the proportion of those the system has labeled the same way. For a set of test items labeled a certain way by the system, *precision* is the proportion of those labeled the same way. For a given label, the *F-measure* is the harmonic mean of precision and recall. The *Receiver Operating Characteristic (ROC)* relates the true positive rate and false positive rate over a varying decision threshold. In engineering a system, one aims to maximize the true positive rate and minimize the false positive rate. For a set of test items of a given label, the true positive rate is the recall. The false positive rate of a given label is the proportion of all test items not labeled that way that the system labels that way. Within the ROC category we include Area-Under-the-Curve, which reduces the ROC to a scalar by integration. Finally, a confusion table (also known as a 'contingency table') notes how many test instances labeled each way are labeled by the system each way.

S1.7 Referencing Critical and Theoretical Work

The following 26 publications are ones we consider are critical of research in MGR. Pampalk et al. [575] notes the inflation of MGR model performance when the confound of artist is not considered in the partitioning of a dataset, and so suggests the use of album and artist filtering. Flexer [572–574] further studies these effects. Other effects from confounding are discovered and studied by Rodriguez-Algarra et al. [405, 481, 569]. Sturm [475, 565, 566] criticizes the use of classification accuracy in MGR experiments, Sturm [471, 568] criticizes the unquestioned use of GTZAN as a dataset for MGR; and Sturm and Collins [477] criticizes the lack of problem definition. Sturm [235, 236, 474, 479, 480] introduces the concept of "horses"—or equivalently "Potemkin Villages".[3] Much more recently, Nie [333] brings into the fold that current MGR experiments are not considering that genre is a dynamic concept, and how such concept drift impacts standard approaches to evaluating MGR systems. More general criticisms of MIR as a research domain and its accepted methodologies are delivered by Flexer [571] (lack of statistical testing), Peeters et al. [576] (challenges in MIR evaluation), Urbano et al. [570] (evaluation validity, reliability, efficiency), Schedl et al. [577] (lack of consideration of users), Sturm [473] (music emotion recognition has the same problems as MGR), and Sturm [482] (research priorities).

Across the corpus we find citations to 24 publications by 17 first authors that we consider to present theoretical perspectives from the social sciences and humanities relevant to work in MGR. These references vary broadly across three dimensions: by their proximity to musical questions, their proximity to genre questions and by discipline. Work that specifically examines musical genre is concentrated in popular music studies (which has historically straddled an intersection of musicology, sociology and cultural studies). From popular music studies, Fabbri's work is perhaps

[3] I. J. Goodfellow *et al.*, "Explaining and harnessing adversarial examples," in *Proc. ICLR*, 2015

most familiar and widely cited in MGR. [587, 588] explore the challenges associated with trying to approach genre relations as hierarchical taxonomies, whilst emphasising the criticality of *relations* to understanding any given genre and the indivisibility of the sonic from the social. More recent—and the current 'state of the art' for musical genre theory—is Brackett's book-length treatment [581], which brings together many threads of genre theory from music and other disciplines to pursue a comprehensive historical survey of how the US music industry used musical and social categories in the 20th century; the opening chapter provides a valuable primer into the many dimensions of musical genre as a topic.

Similar focus on how genres appear and are instrumentalized by the music industry and other stakeholders can be found in papers by Negus [599] and McCleod [594]. Other references to popular music studies work are to more general texts that nevertheless touch on genre [589, 597, 598, 602], and provide useful discussions for ways in which musical genre is signalled beyond the sonic, such as in visual aesthetics. From musicology, two references address genre as a core concern. Drott [586] rebuts, specifically for Western art music, the claim that genre is diminishing in its importance as a mechanism for describing musical difference, whilst Moore [596] confronts distinctions between 'genre' and 'style'. Meyer [595] is a wide-ranging mid 20th Century musicological text theorising the aesthetic role of emotion in music.

Musical genre also receives some attention from sociologists, particularly in the quantitative tradition around the sociology of taste that follows the work of Pierre Bourdieu, and which tends to focus on genre and tastes as signals of social differentiation. Looking at musical genre in this vein is the work of Lena [591, 592] that builds up a structuralist theory of genre from an empirical base; Roy, who like Brackett and Negus, focuses on the role of institutions in propagating genres [600], as well as considering sociological aspects of music more generally [601]; and Coulangeon [582, 583] who considers tastes for musical genres as signals of social stratification.

Work in this sociological tradition also considers genre and taste beyond the musical, such as Lizardo [593] who considers genres and audiences as networks, and DiMaggio [585] who formulates a set of dimensions at work in social classifications of art more widely. Also contending with genre beyond music are Frow [590], whose focus is literature, and Coutinho [584], who considers genre from the perspective of linguistics, again drawing heavily on literary theory. Finally, Bowker and Star [579]—who come respectively from science and technology studies and from sociology—consider classification in general as a cultural phenomenon.

S1.8 Examples of Motivation for MGR

As an example of motivation by industrial need, Martel et al. [303] begins: "In the last decade, automatic music genre classification has become more important as the digital entertainment industry developed". As an example of motivation by public good, Astefanoaei and Collignon [35] begins: "Music genre is the most popular way to describe music—whether in the context of describing one's listening preferences, or when organising music libraries for efficient user access, exploration and discovery of new music". As an example of information overload, Oramas et al. [341] begins: "The advent of large music collections has posed the challenge of how to retrieve, browse, and recommend their contained items". As an example of MIR referential, Rapaso et al. [399] begins: "Music summarization has been the subject of research for at least a decade and many algorithms that address this problem, mainly for popular music, have been published in the past".

S2 Details about GTZAN

When it comes to the index of GTZAN created by Sturm, the URLs pointed to in [471, 475, 568] are no longer available. Sturm's index appears in other locations, however: https://code.soundsoftware.ac.uk/projects/confint/repository/entry/metadata/GTZANindex.txt, and apparently the most up-to-date https://github.com/boblsturm/GTZAN.

When it comes to the fault-filtered partition of GTZAN created by Kereliuk et al. [235, 236], the URL pointed to in that work is no longer available. The fault-filtered partition of GTZAN appears in other locations, however: https://code.soundsoftware.ac.uk/projects/scatter-analysis/repository/show/mca/dataset/folds and https://github.com/boblsturm/GTZAN.

The original website for the GTZAN dataset is http://marsyas.info/downloads/datasets.html, but is inaccessible as of Apr 3 2024. For details on its contents then, we refer to the March 30 2022 snapshot of it by the Internet Archive: https://web.archive.org/web/20220330025414/http://marsyas.info/

downloads/datasets.html. This dataset has multiplied online, in both its original form as digital audio recordings, but also as features. Table 1 shows the URLs we have found for downloading GTZAN, and whether they contain any mention of the faults of the dataset.

URL	Available?	Mentions Faults?
http://marsyas.info/downloads/datasets.html	N	N
https://web.archive.org/web/20220330025414/http://marsyas.info/downloads/datasets.html	Y	N
https://www.kaggle.com/datasets/andradaolteanu/gtzan-dataset-music-genre-classification	Y	N
https://huggingface.co/datasets/marsyas/gtzan	Y	N
https://www.tensorflow.org/datasets/catalog/gtzan	Y	N
https://mirdata.readthedocs.io/en/0.2.0/_modules/mirdata/gtzan_genre.html	Y	N
https://github.com/mir-dataset-loaders/mirdata/tree/master	Y	N
https://datasets.activeloop.ai/docs/ml/datasets/gtzan-genre-dataset	Y	N
https://www.kaggle.com/datasets/carlthome/gtzan-genre-collection	Y	Y
https://music-classification.github.io/tutorial/part2_basics/dataset.html	Y	Y

Supplementary Table 1: GTZAN online locations